RAQIB · FULL WIRING DIAGRAM

Budget robot (under RM 500) · every part and every wire · physical Pi pin numbers · a dot = connected, crossing lines without a dot are not connected

cut USB cable: XL4015 OUT → Pi USB-C power (5.1 V)

Battery 2 × 18650 · 7.4 V

18650 3.7 V +

18650 3.7 V +

switch

XL4015 → 5.1 V

XL4015 IN+ IN- OUT+ OUT- set 5.1V

1000 µF

HUB BOARD (perfboard) · power and I²C rails

5V

3V3

GND

SDA

SCL

Raspberry Pi 4 (2 GB)

40-pin GPIO

ETH

DISPLAY

CPU

RAM

USB 3

USB 2

USB-C power

micro-HDMI

CAMERA

audio

microSD card slot is underneath, on this short edge

3.5 mm cable from the Pi audio jack

PAM8403 amplifier

3 W 4 Ω speaker

USB → Pi USB 3

USB webcam + mic

RIGHT motor + wheel

TT motor

battery + → L298N +12V

LEFT motor + wheel

TT motor

L298N motor driver

OUT1 OUT2 OUT3 OUT4

L298N

5V-EN

+12V GND +5V

ENA IN1 IN2 IN3 IN4 ENB

battery divider (hub)

10k

4.7k

Raspberry Pi 4

the header, enlarged ↓

HC-SR04 CENTRE

HC-SR04

VCC Trig Echo Gnd

1k

2k

HC-SR04 LEFT (45° left)

HC-SR04 RIGHT (45° right)

pin 33 → ENA · pin 35 → ENB (speed) · pins 11/13/15/16 → IN1-IN4 · pin 9 → GND

MPU6050 (0x68)

MPU GY-521

VCC GND SCL SDA XDA XCL ADO INT

AHT20 + BMP280

AHT20+BMP280

VIN GND SCL SDA

ADS1115 (0x48)

ADS ADS1115 16-bit ADC

VDD GND SCL SDA ADDR RT

MQ-135 gas sensor

MQ-135

VCC GND DO AO

10k

20k

LEFT encoder

slot

RIGHT encoder

slot

LM393

VCC GND DO AO

MQ-135 divider (on the hub)

40-pin header

(enlarged)

GND 5V

Trig straight to the Pi · Echo (5 V) through 1k/2k to the Pi

5V 3V3 GND SDA SCL

5 V · 3.3 V · GND · I²C SDA · I²C SCL · battery 7.4 V · PWM speed · motor direction · sonar centre · sonar left · sonar right · encoders · analog / divider · USB cable